4 **B** Ball A:

$h = \frac{1}{2} g t^2 \rightarrow$ time of flight $t = \sqrt{\frac{2h}{g}}$

$$x_A = (2v)t = 2v\sqrt{\frac{2h}{g}}$$

Ball B:

$2h = ½\, gt^2 \rightarrow$ time of flight $t = \sqrt{2}\sqrt{\frac{2h}{g}}$

$$x_B = vt = v\sqrt{2}\sqrt{\frac{2h}{g}}$$

$$\frac{x_A}{x_B} = \frac{2}{\sqrt{2}} = \frac{\sqrt{2}}{1} = 1.41$$

5 A Consider the whole system